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Rollei

BLACK MAGIC KIT

FLÜSSIGE FOTOEMULSION

OPERATING INSTRUCTIONS



IT'S MAGIC

YOUR PHOTOGRAPHIC IDEAS ON THE MOST UNUSUAL MATERIALS

VERWENDUNG IN DER DUNKELKAMMER

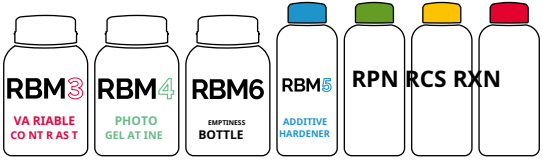
ROLLEI BLACK MAGIC

ROLLEI BLACK MAGICis probably the most comprehensive product system of its kind and therefore offers advantageous features:

- The use of**BLACK MAGIC**should be done in a darkroom.
- High sensitivity, high silver content and therefore short exposure times even for large format images as well as exceptionally high maximum density.
- A complete modular system with interchangeable components.
- Two different emulsions: A normal graduated (**RBM2**) and a variable contrast emulsion (**RBM3**).
- **RBM2**is characterized by high edge sharpness and has a cold image tone due to its silver bromide halides.
- **RBM3**has variable contrast control and has a neutral image tone due to its chlorobromide silver halides.
- **RBM4**is a pure sealing photographic gelatin for water-permeable floors.
- **RBM5**is an additive hardener, free of formaldehyde and chromium potassium sulfate due to a new formula and acts as a powerful hardener.
- The entire product system is supported by proven, high-quality photo chemistry that delivers excellent results.
- Long shelf life and exceptional archival properties through the addition of a minimum amount (0.1%) of a powerful fungicide and bactericide (phenol) for long-term preservation.

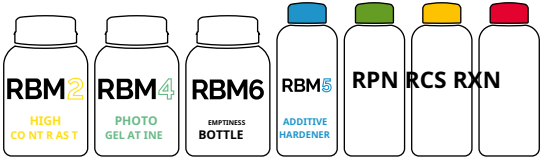
ROLLEI BLACK MAGIC KIT

VARIABLE GRADATION RBM83



300 ml	liquid photo emulsion | Variable gradation | RBM33
100 g	photo gelatin | RBM41
300 ml	empty, light-tight wide-neck bottle | RBM6
250 ml	special hardener | RBM54
250 ml	Developer | RPN12K
250 ml	stop bath | RCS12K
250 ml	pH-neutral fixing bath | RXN12K

NORMAL GRADATION RBM82



300 ml	liquid photo emulsion | Normal gradation
100 g	photo gelatin
300 ml	empty, light-tight wide-neck bottle of
250 ml	special hardener
250 ml	developer
250 ml	stop bath
250 ml	pH-neutral fixing bath

WHAT DO I NEED? A CHECKLIST*

Start with simple surfaces, on which one can practice.

1st What type of material do you want to coat?

2nd What is the contrast of the negatives? If they have different qualities, then choose **RBM3**. If the contrast negatives are fairly consistent, then choose **RBM2**.

3. For most coating applications, the brush application method proves to be the best.

A broad natural hair brush (5 - 10 cm)

4th Have small measuring cups for quantities of 20 – 50ml ready. They are needed to measure small quantities **RBM5** to add.

5th For coating metal, plastic, glass or ceramic tiles, a pre-coating with alkyd resin primer or oil-based polyurethane paint should be carried out; these special primers are available in art stores.

6th For absorbent materials such as textile fabrics, paper or wood, it is recommended to use pure photo gelatine **BLACK MAGIC RBM4** pre-coating.

7th For liquefaction, a bucket of warm water (40 – 45 °C/ 100 – 115 °F) is required.

8th For liquefying and storing small quantities, the **empty BLACK MAGIC wide-neck bottle RBM60** needed.

9th To keep larger quantities liquefied (e.g. to coat large areas or several objects), the large, empty **BLACK MAGIC special bottle RBM65** needed.

10th To improve durability **BLACK MAGIC Additive Hardener RBM5** needed.

11th To facilitate the coating and to minimize the problem of bubbles, spirit (alcohol) is required.

12th Check the safelight before coating to avoid unwanted hazing.

13th Have enough light-tight containers available to store materials containing light-sensitive emulsion.

* Depending on the project, all necessary components should be ready. That is why organization before the coating session is very important.

COATING MATERIALS

Depending on the chosen surface, the thoroughness of the preparation is crucial for the work of art. All materials must be free of dust and grease. If in doubt, clean twice; every trace of grease leads to adhesion problems. Once the material to be coated has been selected, there are three different procedures for the respective material groups:

1st *Absorbent materials such as fabrics, paper or wood:* If you want to prevent the emulsion from penetrating the substrate, a pre-coating with **BLACK MAGIC RBM4**–pure photographic gelatine can be produced (also called baryta solution, but without barium sulfate).

2nd *Smooth and/or hard surfaces such as polished stone, glass or metal:* We also recommend pre-coating with **RBM4** in combination with **RBM5**-Curing additive solution (to improve adhesion and mechanical properties).

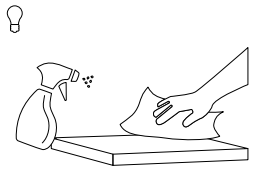
If the emulsion during during the work away- solve, must the underground de with clear alkyd resin primer mer or oil- ized poly- urethane paint (available in art trade) coated. Ver-

3. *All other materials* can be directly connected to the **BLACK MAGIC** Photo emulsion can be coated. For 3D objects, several thin layers ensure a high maximum density (after setting before complete drying).

ON METAL
Due to the electrochemical reaction chain (small electric current between precious and non-precious metals), it is important to apply an insulating layer to the metal surface before coating the light-sensitive emulsion. This can be done using either a white lacquer paint or a clear alkyd paint. Make sure that this primer is completely dry, otherwise the remaining solvent will show in the final result.

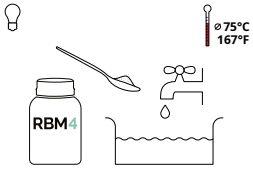
ON PLASTICS
Since all plastics (acrylic, etc.) are manufactured to contain anti-adhesive agents, the emulsion will be stripped during processing. Therefore, it is necessary to coat these materials with paint that will adhere to the emulsion, withstand photographic processing and provide good adhesion to the emulsion. Try the above alkyd paint.

ROLLEI BLACK MAGIC
FIRST STEPS

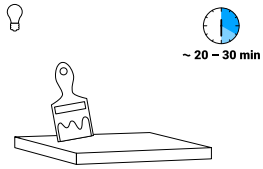


Thoroughly clean the surface to be coated & prepare the photo gelatin RBM4 as follows.

A few drops RWA help the surface tension to reduce.

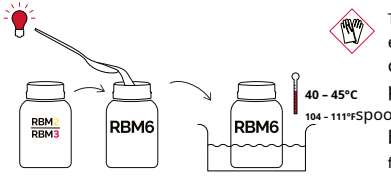


Approach**RBM4**:
Mix 1000 ml distilled water (70 – 80°C/158° – 176°F) + 40 – 80g RBM4 while stirring constantly (40g for non-absorbent | 80g for absorbent surfaces).



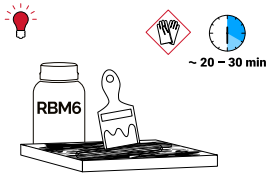
Coat the surface with the photo gelatine and allow it to dry thoroughly for about 20 – 30 minutes.

FROM NOW ON: Wear gloves and safe make sure that darkroom conditions rule.

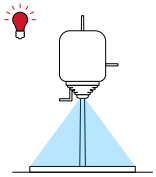


The viscous photo emulsion**RBM2** or**RBM3**with a plastic spoon into the void bottle RBM6 fill. The emulsion sion should be on 40°C. Only use as much as is needed to coat the surface.

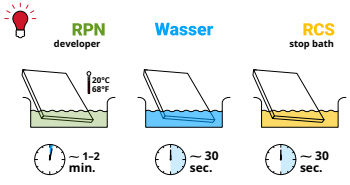
A notice: **Only dark red light use when working with light-sensitive photo emulsion.**



After heating the emulsion, the surface can be coated with a brush. Allow to dry for about 20 - 30 minutes. Depending on which surface is to be exposed, some "test strips" are required for test developments.



To find out which exposure times are best
To ensure the best results, the test strips must first be exposed and developed.



Development:

1000 ml WORKING SOLUTIONS:

RPN+ RBM5
about 1000 ml:
150ml / 175 ml RPN
+ 25 ml / 50 ml RBM5
+ 775 ml water (20°C)

RCS
1+19 =50 ml RCS
+ 950 ml water (20°C)

RXN
1+4 =200 ml RXN
+ 800 ml water

RWA
1+100 =1 ml RWA
+ 999 ml distilled water

PLEASE OBSERVE: Too much RBM5 can lead to a yellow tint in image lead.

LET IT DRY THOROUGHLY AND ENJOY THE UNIQUENESS OF THE ARTWORK!

A NOTICE:
With a spray pistol can Viscosity with distilled Water up to a relationship diluted 1:1 be. must have several layers be sprayed.

COATING & PREPARATION

Surfaces can be coated in a variety of ways:

- **POURING & SWIVELING:** Flat and rigid materials such as tiles, precious metal plates, etc.
- **POURING AND COATING:** Almost all materials
- **DIVE IN:** Smaller three-dimensional objects
- **SPRAYING:** Large and complex structured objects (only use spray guns with a large nozzle opening, an airbrush is not suitable, wear protective clothing/mask, avoid inhaling spray particles).

For most applications, best results are achieved with a brush made of natural hair (eg Japanese brush). We recommend coating test strips at the same time to determine the correct exposure later. Preferably made of the same material or the back of photo paper).

COATING:

The cooling and curing phase follows immediately after coating. At this stage, the coated surface is very delicate. The setting takes about 1 – 2 hours. The setting can be accelerated by keeping the coated materials in the refrigerator for about 30 minutes. To avoid unwanted light influence, either use a light-tight container or unscrew the light bulb of the refrigerator.

DRYING:

Now the coated materials can dry. You can use a hairdryer at the lowest temperature setting and at a medium distance to avoid melting the emulsion. Check the hairdryer beforehand by turning it on in the dark to make sure there is no electrical spark or the glow of the heating wires that could cause unwanted exposure. Of course, the coated objects can also be dried in the darkroom (but make sure the lights are off!). This takes between a few hours and a day, depending on the type of object and the humidity.

EXPOSURE

The process is similar to that of standard photographic papers, especially when coated flat materials are used. For three-dimensional objects, the lens aperture must be closed to the smallest value to achieve maximum depth of field (when adjusting the focus, remember that stopping down extends the depth of field approximately 1/3 in front of and 2/3 behind the set plane of focus).

Tip: With cylindrical or evenly shaped objects, a large halftone negative (or if you want a graphic negative without gray values) can be created using "GENIUS Litho Film". This negative can then be applied to the coated object (eg a bottle or vase) and then rotated evenly under the enlarger light - this makes sharp and even exposure much easier!

Becomes **RBM3** used for coating, the gradations can ensure more optimal results. As with conventional photo papers with variable contrast (eg ROLLEI Vintage 111), the gradation can be regulated either with color filters for the filter drawer (ILFORD Multigrad Filter/Foma Filter) or with color heads/enlargers (yellow - magenta). This is possible within the gradations No.1 (soft) to No.4 (hard).

Yellow = for a softer image reproduction

Magenta = for a harder image reproduction

If the exposure was too strong, it can be bleached with ROLLEI RST SEPIA PART A, dilution 1+30, to the desired density.

DEVELOPMENT

The**ROLLEI-Chemie**is especially good for**BLACK MAGIC**Keep the operating temperature to max. 20°C or lower to avoid damaging the emulsion.

DEVELOPER:Rollei RPN for normal contrast; Rollei RHC for higher contrast.

SHORT WATERING: Rinse with tap water or soak briefly in a bowl for 10 – 20 seconds.

STOPPBAD:Rollei RCS, odorless, environmentally friendly, citric acid based.

FIXER:Rollei RXN, pH-neutral for low-acid fixation & archival strength, avoids sulphur dioxide emissions, shorter washing times.

WATERING: Tap water, approx. 20°C/68°F, duration depends on the thickness of the coating & the substrate; at least 15 min.

WETTERING AGENTS:Rollei RWA, with bactericidal & fungicidal additives.

AFTER-TREATMENT:Rollei SELENIA, high-purity selenium toner for extreme image silver protection & increased brilliance.

DRYING: Air dry, use hair dryer if necessary.

TIPS FOR OPTIMAL PROCESSING:

1st Avoid influences on the coated emulsion, even if curing has taken place, the emulsion is very sensitive.

2nd For larger objects, place them on a sufficiently large tray/in a larger bowl and carefully pour the chemical over them/gently immerse the object. Proceed in the order above.

3.Hardening fixative baths must not be used, otherwise long washing times will dissolve the emulsion.

4th Larger objects (room walls, etc.) that cannot be treated with the means described: Pour chemicals onto the object from bottom to top using a watering can. The procedure is identical to the steps described previously.

5th Always use fresh fixer and keep the fixation time as short as possible to achieve shorter wash times.

GOOD TO KNOW

1st Da**BLACK MAGIC**is a very versatile product, our application information should be followed carefully for the time being.

2nd Finished objects can be protected by coating them with a clear polyurethane paint, for example.

3.To avoid an electrochemical reaction, metal objects to be coated should be coated with a primer (**RBM4 PHOTO GELATINE**) must be pretreated.

4th**BLACK MAGIC ADDITIVE HARDENER RBM5**is formaldehyde-free and can also be used for emulsion hardening of films and papers. For this, use 40 – 80 ml**RBM5**Dissolve in 1000 ml of water and immerse the materials to be hardened for 2 to 3 minutes and then allow to dry.

TROUBLESHOOTING

RESULT	REASON	RESOLUTION
blisters in the finished image	Shaking/Stirring has air bubbles caused	Strong stirring/ Avoid shaking 10–20ml of spirit 500ml RBM at slow menStirAdd
Uneven coating	inclined storage before/during curing phase	Base with a spirit level align
detachment of the emulsion during processing	fats/anti-stick agents on the object	To be coated material thoroughly clean, if necessary pre-treat with alkyd resin paint

FREQUENTLY ASKED QUESTIONS

How much area can be coated?

For non-absorbent substrates, up to 8 m2, for suction up to 4 m2In practice, this may vary depending on the thickness of the coating and the absorption properties of the material chosen.

How do I store BLACK MAGIC?

The liquid emulsions**RBM23**and**RBM33**should be stored in the refrigerator at approx. 6 – 8°C / 40 – 46 F. All other ingredients, such as**RBM4**and**RBM5**can be stored at room temperature.

How long is the shelf life?

If the chemical is stored under the above conditions, the shelf life is about one year. If**RBM2**and**RBM3**At temperatures below -10 C°/14°F the shelf life is significantly extended.

What type of Safelight is suitable?

RBM2:**dark red**

RBM3:**dark red**

Silver halides are light sensitive and will expose even the smallest amounts of light. It is therefore advisable to expose the coated objects to as little safelight as possible. In reality, no safelight is truly safe - it is just a question of the length of time the light is exposed to.

We recommend a *veil test* under your own conditions through-supply – this is particularly important if large areas/ many objects coated want to become.

How do I do a veil test?

Perform a test coating and partially expose the area where all other materials are to be coated to the safelight. Place a piece of black card on the object and move it for a series of exposures of 15 minutes/30 minutes/45 minutes. Develop the test object and determine if haze is noticeable. If so, move the safelight further away from the work area or replace the safelight.

How do I achieve different image tones?

If with**RBM4**If a pre-coating is done, water-soluble photo dyes/pigments can be added to the solution - as much as necessary to achieve the desired color tone, usually a few drops are sufficient.

How do you combine the modular components of**BLACK MAGIC**?

The solid-graduated emulsion**RBM2**can be mixed in any ratio to achieve different gradations. For a wide range of different contrasts, we recommend using**RBM3**. Absorbent materials should be cleaned with the purified photo gelatine**RBM4**pre-treated – do not use gelatine for food!**additive hardener RBM5** provides improved resistance. Due to the special formula,**RBM5**also be added to the emulsion itself and the developer.**RBM6**It is easy to warm up portions of the emulsion, the special bottle is light-tight.

How do I prepare a barytage solution?

Heat 1l of tap water to 70–80°C and add 40g**RBM4** Dissolve completely by continuous stirring. Reduce the surface tension by adding a few drops**ROLLEI WETTING AGENT**(facilitates the even coating of smooth/hard materials). Coat the material immediately afterwards. To improve the curing of the emulsion, add 20 – 50 ml**RBM5**add to the solution.

HARDENING NOTE
Indirect hardening of the barite solution by adding**RBM5**. Direct hardening by adding**RBM5**to the light-sensitive photo emulsion (in this case please use the photo emulsion within 2 hours). To improve the coating adhesion and the resistance of the surface, both methods can be combined. In addition,**RBM5**be added directly to the developer.

The**BLACK MAGIC**Remove the photo emulsion from the refrigerator and carry out the following steps under safe light(!)carry out:

Either with a plastic or stainless steel spoon, pour the required amount into the special bottle**RBM6**and liquefy in a container filled with warm water (35 – 40°C). | For larger application projects, heat the original bottle using the same means.

*To change or dilute the consistency of the light-sensitive layer, do not stir distilled water into Black Magic! This would have a lasting negative effect on adhesion. For a barytage solution, use pure**BLACK MAGIC PHOTO GELATINE RBM4**In this case, only the silver content is reduced and the coating is ideal for several applications.*

If larger objects (e.g. ceramics, stones, etc.) are to be coated, they can be heated in the oven (30 – 40°C /86 – 104°F) to prevent the emulsion from hardening during coating.

For initial tests or for smaller objects to be coated, simply remove the required amount from the original bottle (under a darkroom safelight) and pour it into the wide-mouth RBM6 bottle included in the kit. Then pour this into a container (or bucket) filled with warm water to liquefy. This will ensure the quality of the rest of the BLACK MAGIC, as repeated heating and cooling will damage light-sensitive products.

CONTACT

If you have any questions or suggestions after reading this manual, please send them to the following address:

HANS O. MAHN GmbH & CO. KG · MACO PHOTO PRODUCTS
Brookstieg 4 | 22145 Stapelfeld | Germany
www.rolleianalog.com

SECURITY

In all areas of life, any product can be harmful if used improperly. This of course also applies to photographic products. At MACO PHOTO PRODUCTS, we care about our customers and our environmental responsibilities. That is why we have reduced the possible health risks to a minimum. However, we ask that you exercise due caution when handling our photographic products.

- **Always keep the products out of the reach of children!**
- **Read these instructions carefully and attentively**
  – in case of doubt, we are always available to provide advice and assistance.
- **Work with photochemistry should be carried out in rooms with good ventilation.**
- **We also provide safety data sheets.**

DISPOSAL

For information on the disposal of photochemicals and their used solutions, please contact your local environmental officer. European Waste Catalogue:

Developer 09 01 01 01 01 01

Water-based developer and activator solutions

Bleach-fixing bath 09 01 05 05

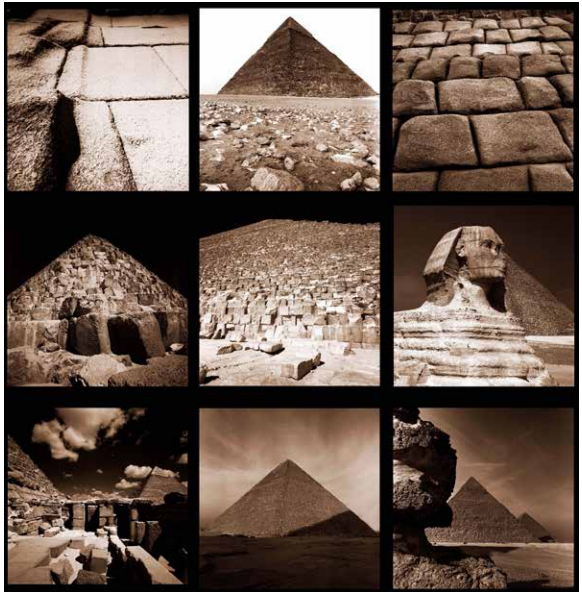
bleaching solutions and bleach-fixing baths

Bottles and canisters are increasingly being recycled. In both cases, it is important that the containers do not contain any concentrate residues. We therefore recommend rinsing out concentrate residues with some of the batch water and adding it to the respective batch.

ART WITH ROLLEI BLACK MAGIC BY KAROLINE AMAURY



Karoline Amaury, born in 1963 in Kortrijk (Flanders, Belgium), lives and works as an artist in Italy and Egypt. She used photography to study the lighting scenarios in and around holy places and sacred architecture. She has been working with Rollei quality products for 20 years. Under the name "Ta-Mery", one of the names of ancient Egypt, she has been making her photographic excursions through Egypt for 13 years. Karoline Amaury uses the medium of photography not only for a representational purpose, but also uses a variety of special photo techniques and cameras to enable different possibilities of representation and discussion.



Egypt Heliopolis – Pyramids, Rollei Black Magic on paper & sepia toning,
1m × 1m, 2018



Portrait of Ramses II
Luxor Temple Egypt,
Emulsion on alabaster,
50 × 70 cm, 2016

Detail of Horus
Temple of Dendera Egypt,
emulsion on piece of wood,
40 cm, 2016



NOTES





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